

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

ASSIGNMENT-1

DURATION: 3 WEEKS

SUMMER SEMESTER, 2023-2024

FULL MARKS: 15

CSE 4621: Microprocessor and Interfacing

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 1 (one)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

1. Design a project using a microcontroller (e.g., Arduino) or microcomputer (e.g., Raspberry Pi), to solve a real-world problem (e.g., in IUT or Bangladesh). The project design should include:

(A) Problem Statement:

- What are you planning to solve?
- Why do you need to solve it?
- How are you planning to solve it?

(B) Technical Specifications:

Enlist the items you need for the project, e.g.,

- **IR Sensor:** Detects and measures infrared radiation.
- **Wires:** Connects components electrically.

(C) Design Diagram:

Design a diagram showing the solution setup. The diagram should include I/O devices, interfaces, and processing units. Include abstract entities if required.

Bonus Points:

1. (+2) Include model types/number and a cost breakdown of the project (in BDT).
2. (+3) Create the report in $\text{\LaTeX}$.
3. (+10) Implementation of the proposed solution. Include a video demonstration of the project.
Can be shown after the 3-week deadline.

Instructions for submission:

1. Submit a single document containing the design report. *Please don't submit scans of hand-written assignments.*
2. Please keep the assignment concise. Avoid unnecessary or over-complicated AI-generated sentences. The problem statement should be within one short paragraph, and the technical specification should be one line for each item.
3. Recommended tools for the diagram: draw.io, Lucidchart.
4. Plagiarism and similarity will be checked. Ensure that your assignment has not been shared with anyone. In cases of high similarity, both assignments will be penalized.
5. Fully AI-generated assignments will be penalized. However, AI can be used to do research or refine the writing.